

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: PCMS_ET

robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

for simulation

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#	Component	Part name	Autor
1	PCB with electronic components	Custom made	Name of team members/Jlcpcb
2	Wheels	Nexus Robot Mecanum Wheel	Nexus Robot
3	Sheet metall parts for frame	Custom made	
4	Drive motors	Maxon ECX TORQUE 22 XL Ø22 mm	Maxon Group
5	Robots shell	Custom made	Name of team members
6	Robots costume	Custom made	Name of team members
7	Microcontroller	Arduino Mega 2560	Arduino
8	Connecting parts	Screws, nuts, washers, ...	
9	Motors for arm	NEMA-17-01 NEMA-17-01 0.4 Nm 1.68 A	Conrad Electronic
10	Stepper motor controllers	Pololu Tic T825	Electronics
11	LEDs	Various colours and sizes	Company
12	Battery	11.1 V 3700 mAh	Company
13	Cables	Various cables	
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1	PCB with electronic components (MAIN BOARD)	Custom made	PCMS
2	PCB with electronic components(SUB BOARD)	Custom made	PCMS
3	Wheels	Custom made	
4	Robots shell and frame	Custom made	
5	Microcontroller	ESP32	espressif
6	Connecting parts	Screws, nuts, washers, ...	Company
7	Motors for arm	Servo SG90	Company
8	Drive motors	25GA-370 12V 58RPM DC Reducer Gear Motor with Encoder	Farnell An Avnet Company

9	Drive motors	Motor	Company
10	LED	Custom made	Company
11	Gyro	JY61P	Company
12	Pressure Sensor	pressure sensor	Company
13	laser ranging sensor	VI53LOX	Company
14	camera	Maixcam	Company
15	motor controllers	TB6612FNG	Company
16	Battery	18650 3.7V 12000mwh	Company
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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Brazilian Real	R\$11,824.09	\$2,306.39

Source	Quantity	Unit cost (Local)	Unit cost (USA Dollars)	Total Cost (Local)	Total Cost (USA Dollars)
jlcpcb.com	1	R\$338.80	\$60.00	R\$338.80	\$60.00
robotshop.com	4	R\$112.93	\$20.00	R\$451.72	\$80.00
Local workshop	1	R\$1,129.34	\$200.00	R\$1,129.34	\$200.00
Maxon Motors	4	R\$1,694.01	\$300.00	R\$6,776.04	\$1,200.00
3D Printer	1	R\$112.93	\$20.00	R\$112.93	\$20.00
Self made	1	R\$282.34	\$50.00	R\$282.34	\$50.00
conrad.de	2	R\$237.16	\$42.00	R\$474.32	\$84.00
Local tool store	1	R\$225.87	\$40.00	R\$225.87	\$40.00
conrad.de	2	R\$118.58	\$21.00	R\$237.16	\$42.00
conrad.de	2	R\$508.20	\$90.00	R\$1,016.40	\$180.00
farnell.com	20	R\$2.82	\$0.50	R\$56.40	\$10.00
farnell.com	3	R\$203.28	\$36.00	R\$609.84	\$108.00
Local electronics store	1	R\$112.93	\$20.00	R\$112.93	\$20.00
jlcpcb.com	1		USD 0.74		USD 0.74
jlcpcb.com	1		USD 0.74		USD 0.74
3D Printer	4		USD 1.00		USD 1.00
3D Printer	1		USD 1.00		USD 1.00
https://www.espressif.com/en/products/socs/esp3	2		USD 11.00		USD 22.00
taobao.com	1		USD 1.00		USD 1.00
taobao.com	1		USD 0.80		USD 0.80
Yixing Motors	2		USD 3.20		USD 6.40

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#	Name	Software's Tool/Library	Description
1	VL53L0X-Arduino	Library	For controlling the VL53L0X distance sensor on Arduino
2	Open CV	Library	Image processing library
3	IDE Arduino	Software's Tool	Software
4	Visual Studio Code	Software's Tool	Software
5	Numpy	Library	Array and math library
6	Matplotlib	Library	Python graphics generation library
7	Tinckercad	Software's Tool	Online platform for simulating circuits
8	Git	Software's Tool	Version control for managing functions
9	Fusion 360	Software's Tool	3D CAD shell design
10	Tic Arduino	Library	Library for controlling a Nema 17 stepper motor with the Tic T825 driver
11	Modulemotorl298n	Library	L298N module without having to worry about the pins manually in each
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#	Component	Part name	Autor	Source	Quantity	(Local
1	components (MAIN BOARD)	Custom made	PCMS	jlcpcb.com	1	
2	components(SUB BOARD)	Custom made	PCMS	jlcpcb.com	1	
3	Wheels	Custom made	PCMS	3D Printer	4	
4	Robots shell and frame	Custom made	PCMS	3D Printer	1	
5	Microcontroller	ESP32	espressif	https://www.	2	
6	Connecting parts	Screws, nuts, washers, ...	Company	taobao.com	1	
7	Motors for arm	Servo SG90	Company	taobao.com	1	
8	Drive motors	Gear Motor with Encoder	Company	Yixing Motors	2	
9	Drive motors	Gear Motor	Company	Yixing Motors	2	
10	LED	Custom made	Company	electronic	1	
11	Gyro	JY61P	Company	WIT	1	
12	Pressure Sensor	pressure sensor	Company	Technology	2	
13	laser ranging sensor	VL53LOX	Company	electronic	10	
14	camera	Maixcam	Company	m	2	
15	motor controllers	TB6612FNG	Company	electronic	2	
16	Battery	18650 3.7V 12000mwh	Company	geshiyun	3	
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1	VL53L0X-Arduino	Library	For controlling the VL53L0X distance sensor	pololu/vl53l0x-arduino: Biblioteca Pololu Ar
3	Arduino Ide	Software's Tool	Software	Software Arduino
4	Thonny	Software's Tool	Software	Software Thonny
5	MaixVision	Software's Tool	Software	Software Maixcam
6	ONSHAPE	Software's Tool	3D design	onshape.com
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